

How the OCR Service Works

A plain-language walkthrough of what happens after you upload a document — how many calls to Gemini it takes, what each one is for, and where the time actually goes.

In short: every document goes through up to four separate checks with Gemini before it's trusted, and documents uploaded together get one extra check comparing them to each other. Today, all of this runs one step at a time — nothing happens at the same time as anything else. That is the single biggest reason a multi-document upload takes noticeably longer than a one-document upload, and it is also the easiest thing to improve.

1. The journey of one uploaded document

- 1 File checks** — is it really a JPG/PNG/PDF, not empty, not too large, not corrupted?
no Gemini call
- 2 Extraction** — the document image is sent to Gemini, which reads every field (name, DOB, ID number, etc.) and reports what type of document it actually sees. **Gemini call #1**
- 3 Type check** — does what Gemini actually saw match the document type you selected in the dropdown? If not, stop here.
- 4 Completeness check** — for a National ID / passport / visa, is both the front and back present? If not, stop here.
- 5 Self-verification** — Gemini looks at the same image a second time, independently, and is asked to judge it: is it genuine, legible, internally consistent, not expired? **Gemini call #2**
- 6 If anything above found a problem**, the whole request stops right here, for every document in the upload — nothing below this line runs, and no document is cropped or rotated for nothing.

Only if every uploaded document gets this far does the process continue to the next section — comparing documents to each other, then cropping and straightening them.

2. Once every document has individually passed

Cross-verification — only when 2 or more documents were uploaded together. Gemini is shown the extracted details from all of them at once and asked to find anything that disagrees — a different name, a different date of birth, a mismatched number. This is one single call for the whole batch, not one per document. **Gemini call, batch-wide**

If it finds a disagreement, the whole request stops here too — again, before anything is cropped or rotated.

Cropping — for each document, Gemini locates exactly where the document is in the photo/scan so it can be cut out cleanly. **Gemini call, per document**

Rotation — for each document, Gemini looks at the cropped result and decides which way is "up" so it can be straightened. **Gemini call, per document**

Result — every document's extracted fields are merged into one single JSON answer.

3. How many Gemini calls does this actually add up to?

Stage	How often it runs	Runs per...
Extraction	Always, for every document	document
Self-verification	If validation is switched on — but only if the document already passed the type and completeness checks	document
Cross-verification	Only when 2+ documents are uploaded together, and only once every one of them has individually passed	whole batch (one call, not per document)
Cropping	If cropping/rotation is switched on, and only once the entire batch has passed everything above	document
Rotation	Same as cropping	document

Worked example:

Documents uploaded	Best case, all valid
1 document	$1 + 1 + 1 + 1 = 4$ calls (no cross-verification — nothing to compare it to)
2 documents	$(1+1+1+1) \times 2 + 1$ cross-check = 9 calls
3 documents	$(1+1+1+1) \times 3 + 1$ cross-check = 13 calls

(If a document fails a check early, its later calls simply never happen — a rejected document never reaches self-verification or cropping, so the real count is often lower than "best case".)

4. Where does the time actually go? (a real, recent example)

This is an actual request from this service's own logs today — two documents (a National ID and a Passport), uploaded together, both valid, with checking and cropping both switched on. Total time: **26.15 seconds**.



Notice the first two bars: the Passport's own checks did not start until the National ID's checks had *completely* finished — even though the two documents have nothing to do with each other at that stage. That is 12 seconds spent one-at-a-time on two tasks that don't depend on each other at all. The same is true of the final stretch — cropping the National ID, then only afterwards cropping the Passport, one after the other.

5. Can this be made faster? Yes — with one thing to decide

✓ **Safe to speed up, no downside**

A — Check every document at the same time, not one after another.

Each document's own extraction and self-verification only needs that document's own picture — it never needs to know anything about any other document in the upload. There is no reason to make the Passport wait for the National ID to finish first. Running them together instead of in a queue would shrink that part of the time from "the sum of every document" down to roughly "however long the single slowest document takes" — for the example above, that would cut ~12 seconds down to ~6.6 seconds.

B — Crop and rotate every document at the same time, not one after another.

By the point cropping starts, every document has already been fully approved — there is nothing left to protect against by doing them one at a time. Running all of them together is a pure time saving with no trade-off at all.

⚖️ **Possible, but needs a decision first**

C — Start a document's self-verification at the same moment as its extraction, instead of waiting for extraction to finish.

Self-verification only looks at the picture — technically it doesn't need extraction's answer to begin. Starting both together would save time. But today, self-verification deliberately only runs *after* we already know the document is the right type and has both sides present — so a document that turns out to be the wrong type, or missing a page, never wastes a self-verification call at all. Starting both at once would mean occasionally paying for and running a self-verification check on a document that gets rejected a moment later anyway. It's a genuine trade-off between "a little faster, occasionally wasteful" and "a little slower, never wasteful" — worth deciding on purpose rather than as a side effect.

— Cannot be sped up — and that's expected

Cross-verification has to wait for every document. Comparing documents to each other is only possible once every one of them has already been individually checked — there's nothing to compare until then. This step is a genuine, unavoidable checkpoint, not an inefficiency.

6. Bottom line

4 possible Gemini calls per document

+1 shared call when 2+ documents are uploaded together

everything currently runs one step at a time

Two of the improvements above (**A** and **B**) are pure wins — they would noticeably cut down total waiting time for anyone uploading more than one document, with no change in cost, accuracy, or behaviour. The third (**C**) is faster still, but only makes sense once there's a clear answer to "is it worth occasionally paying for a check we end up throwing away, in exchange for extra speed?"

Generated from the service's own code and its real request logs on 2026-09-24. No live requests were made to produce this document.